HW1: SIR and SEIR Model Implementation.

Initial Hypothesis.

B: Infection Peak in the SIR Model

I think the number of the infected people will firstly increase and then decrease. I expect the peak to happen around day 70-100, about 400-600 people. Since R0 is greater than 1, the disease can spread at the begining. But, as more people get infected and recover, the number of susceptible people will become smaller. Therefore, I expect the number of infected people will be decreasing after reaching the peak.

D: ~~Epidemic~~ Epidemic Waves in the SEIR Model.

I expect multiple waves of infection especially in the longer simulation. The model inlludes a delay before infected people become infectious. At the same time, new births add susceptible people over time. Because of these factors, the infetion may increase again after the first wave in my opnion.

E: Sensitivity Analysis.

In my opnion, the transmission rate β will affect the infection peak more than the recovery rate γ. A higher β makes the disease spread faster and may lead to more infections at the same time. Although a higher γ helps people to recover faster, I expect β to have a stronger affect on the peak in this experiment.